

Table 1: LSDA improvement over baselines for configuration dependency *identification*.

| System           | cDep                  | GPTuner               | CoT                   | LLM-RAG               |
|------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                  | Mean (improvement)    | Mean (improvement)    | Mean (improvement)    | Mean (improvement)    |
| <b>Precision</b> |                       |                       |                       |                       |
| HDFS             | 0.125 (178.4%)        | 0.048 (625.0%)        | 0.097 (258.8%)        | 0.235 (48.1%)         |
| HBASE            | 0.094 (351.1%)        | 0.029 (1362.1%)       | 0.059 (618.6%)        | 0.203 (108.9%)        |
| YARN             | 0.406 (34.0%)         | 0.086 (532.6%)        | 0.111 (390.1%)        | 0.250 (117.6%)        |
| MAPREDUCE        | 0.141 (175.9%)        | 0.062 (527.4%)        | 0.147 (164.6%)        | 0.283 (37.5%)         |
| CORE             | 0.110 (391.8%)        | 0.036 (1402.8%)       | 0.073 (641.1%)        | 0.243 (122.6%)        |
| GLANCE           | —                     | 0.075 (532.0%)        | 0.095 (398.9%)        | 0.303 (56.4%)         |
| CINDER           | —                     | 0.056 (548.2%)        | 0.037 (881.1%)        | 0.091 (298.9%)        |
| <b>Average</b>   | <b>0.175 (151.4%)</b> | <b>0.056 (685.7%)</b> | <b>0.088 (400.0%)</b> | <b>0.230 (91.3%)</b>  |
| <b>Recall</b>    |                       |                       |                       |                       |
| HDFS             | 0.519 (27.9%)         | 0.429 (54.8%)         | 0.729 (-8.9%)         | 0.369 (79.9%)         |
| HBASE            | 0.455 (90.8%)         | 0.460 (88.7%)         | 0.369 (135.2%)        | 0.598 (45.2%)         |
| YARN             | 0.608 (54.6%)         | 0.354 (165.5%)        | 0.607 (54.9%)         | 0.472 (99.2%)         |
| MAPREDUCE        | 0.422 (110.9%)        | 0.244 (264.8%)        | 0.530 (67.9%)         | 0.419 (112.4%)        |
| CORE             | 0.155 (427.7%)        | 0.080 (922.5%)        | 0.553 (47.9%)         | 0.541 (51.2%)         |
| GLANCE           | —                     | 0.064 (1326.6%)       | 0.155 (489.0%)        | 0.268 (240.7%)        |
| CINDER           | —                     | 0.090 (887.8%)        | 0.245 (262.9%)        | 0.218 (307.8%)        |
| <b>Average</b>   | <b>0.432 (97.9%)</b>  | <b>0.246 (247.6%)</b> | <b>0.455 (87.9%)</b>  | <b>0.412 (107.5%)</b> |
| <b>F1-score</b>  |                       |                       |                       |                       |
| HDFS             | 0.202 (125.7%)        | 0.086 (430.2%)        | 0.170 (168.2%)        | 0.287 (58.9%)         |
| HBASE            | 0.156 (237.8%)        | 0.055 (858.2%)        | 0.101 (421.8%)        | 0.300 (75.7%)         |
| YARN             | 0.487 (41.3%)         | 0.138 (398.6%)        | 0.187 (267.9%)        | 0.325 (111.7%)        |
| MAPREDUCE        | 0.211 (155.5%)        | 0.098 (450.0%)        | 0.230 (134.3%)        | 0.334 (61.4%)         |
| CORE             | 0.129 (400.8%)        | 0.049 (1218.4%)       | 0.129 (400.8%)        | 0.334 (93.4%)         |
| GLANCE           | —                     | 0.069 (800.0%)        | 0.117 (430.8%)        | 0.281 (121.0%)        |
| CINDER           | —                     | 0.069 (643.5%)        | 0.064 (701.6%)        | 0.127 (303.9%)        |
| <b>Average</b>   | <b>0.237 (140.5%)</b> | <b>0.081 (603.7%)</b> | <b>0.143 (298.6%)</b> | <b>0.284 (100.7%)</b> |
| <b>AUC</b>       |                       |                       |                       |                       |
| HDFS             | 0.758 (9.8%)          | 0.711 (17.0%)         | 0.862 (-3.5%)         | 0.684 (21.6%)         |
| HBASE            | 0.725 (28.7%)         | 0.722 (29.2%)         | 0.681 (37.0%)         | 0.798 (16.9%)         |
| YARN             | 0.803 (20.8%)         | 0.674 (43.9%)         | 0.800 (21.2%)         | 0.735 (32.0%)         |
| MAPREDUCE        | 0.708 (33.3%)         | 0.618 (52.8%)         | 0.762 (23.9%)         | 0.708 (33.3%)         |
| CORE             | 0.577 (57.5%)         | 0.538 (69.0%)         | 0.773 (17.6%)         | 0.769 (18.2%)         |
| GLANCE           | —                     | 0.531 (80.0%)         | 0.577 (65.7%)         | 0.633 (51.0%)         |
| CINDER           | —                     | 0.544 (73.5%)         | 0.617 (53.0%)         | 0.607 (55.5%)         |
| <b>Average</b>   | <b>0.714 (29.8%)</b>  | <b>0.620 (49.5%)</b>  | <b>0.725 (27.9%)</b>  | <b>0.705 (31.5%)</b>  |

Table 2: LSDA improvement over baselines for configuration dependency *classification*.

| System           | cDep                  | CoT                   | LLM-RAG               |
|------------------|-----------------------|-----------------------|-----------------------|
|                  | Mean (improvement)    | Mean (improvement)    | Mean (improvement)    |
| <b>Precision</b> |                       |                       |                       |
| HDFS             | 0.030 (666.7%)        | 0.065 (253.8%)        | 0.133 (72.9%)         |
| HBASE            | 0.019 (926.3%)        | 0.049 (298.0%)        | 0.070 (178.6%)        |
| YARN             | 0.179 (126.8%)        | 0.030 (1253.3%)       | 0.132 (207.6%)        |
| MAPREDUCE        | 0.047 (461.7%)        | 0.112 (135.7%)        | 0.244 (8.2%)          |
| CORE             | 0.025 (1040.0%)       | 0.088 (223.9%)        | 0.147 (93.9%)         |
| CINDER           | —                     | 0.038 (802.6%)        | 0.122 (181.1%)        |
| GLANCE           | —                     | 0.140 (10.0%)         | 0.056 (175.0%)        |
| <b>Average</b>   | <b>0.060 (346.7%)</b> | <b>0.075 (257.3%)</b> | <b>0.129 (107.8%)</b> |
| <b>Recall</b>    |                       |                       |                       |
| HDFS             | 0.196 (150.0%)        | 0.232 (111.2%)        | 0.188 (160.6%)        |
| HBASE            | 0.067 (340.3%)        | 0.097 (204.1%)        | 0.178 (65.7%)         |
| YARN             | 0.377 (58.9%)         | 0.180 (232.8%)        | 0.280 (113.9%)        |
| MAPREDUCE        | 0.106 (261.3%)        | 0.122 (213.9%)        | 0.204 (87.7%)         |
| CORE             | 0.239 (100.4%)        | 0.209 (129.2%)        | 0.229 (109.2%)        |
| CINDER           | —                     | 0.359 (39.8%)         | 0.235 (113.6%)        |
| GLANCE           | —                     | 0.126 (220.6%)        | 0.083 (386.7%)        |
| <b>Average</b>   | <b>0.197 (128.4%)</b> | <b>0.200 (125.0%)</b> | <b>0.200 (125.0%)</b> |
| <b>F1-score</b>  |                       |                       |                       |
| HDFS             | 0.104 (245.2%)        | 0.078 (360.3%)        | 0.161 (123.0%)        |
| HBASE            | 0.057 (682.5%)        | 0.053 (741.5%)        | 0.170 (162.4%)        |
| YARN             | 0.237 (115.2%)        | 0.081 (529.6%)        | 0.154 (231.2%)        |
| MAPREDUCE        | 0.110 (241.8%)        | 0.091 (313.2%)        | 0.248 (51.6%)         |
| CORE             | 0.049 (977.6%)        | 0.068 (676.5%)        | 0.213 (147.9%)        |
| CINDER           | —                     | 0.044 (1190.9%)       | 0.141 (302.8%)        |
| GLANCE           | —                     | 0.092 (220.7%)        | 0.114 (158.8%)        |
| <b>Average</b>   | <b>0.111 (296.4%)</b> | <b>0.072 (511.1%)</b> | <b>0.165 (166.7%)</b> |
| <b>AUC</b>       |                       |                       |                       |
| HDFS             | 0.670 (12.8%)         | 0.670 (12.8%)         | 0.577 (31.0%)         |
| HBASE            | 0.567 (44.3%)         | 0.572 (43.0%)         | 0.632 (29.4%)         |
| YARN             | 0.674 (26.7%)         | 0.646 (32.2%)         | 0.620 (37.7%)         |
| MAPREDUCE        | 0.599 (32.4%)         | 0.594 (33.5%)         | 0.619 (28.1%)         |
| CORE             | 0.544 (54.2%)         | 0.610 (37.5%)         | 0.642 (30.7%)         |
| CINDER           | —                     | 0.580 (60.3%)         | 0.587 (58.4%)         |
| GLANCE           | —                     | 0.545 (42.6%)         | 0.557 (39.5%)         |
| <b>Average</b>   | <b>0.620 (32.9%)</b>  | <b>0.612 (34.6%)</b>  | <b>0.604 (36.4%)</b>  |

Table 3: LSDA improvement over baselines for configuration dependency *formalization* correctness.

| System         | CoT                   | LLM-RAG              |
|----------------|-----------------------|----------------------|
|                | Mean (improvement)    | Mean (improvement)   |
| HDFS           | 12.0% (291.7%)        | 23.0% (104.3%)       |
| HBASE          | 44.0% (43.2%)         | 46.0% (37.0%)        |
| YARN           | 22.0% (181.8%)        | 48.0% (29.2%)        |
| MAPREDUCE      | 42.0% (73.8%)         | 53.0% (37.7%)        |
| CORE           | 39.0% (66.7%)         | 35.0% (85.7%)        |
| CINDER         | 25.0% (184.0%)        | 20.0% (255.0%)       |
| GLANCE         | 29.0% (131.0%)        | 73.0% (-8.2%)        |
| <b>Average</b> | <b>30.4% (110.5%)</b> | <b>42.6% (50.2%)</b> |